Predictive Analytics

PROJECT REPORT

(Project Semester August-December 2025)

BANK LOAN DEFAULT PREDICTION
USING PREDICTIVE ANALYTICS

Submitted by

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B.Tech (CSE) - K23MK

Course Code: INT234

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Vivek Sharma** bearing Registration no. **12314253** has completed **Predictive Analytics (INT234)** project titled, **"BANK LOAN DEFAULT PREDICTION "** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Mr. Jaffar Amin Chacket

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 15/12/2025

DECLARATION

I **Vivek sharma**, student of **B.Tech CSE - K23MK** under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20/12/2025 Signature :

Vivek.

Registration No. **12314253** Name -**Vivek sharma**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide **Mr. Jaffar Amin Chacket** for her valuable guidance, support, and encouragement throughout this project. I am thankful to **Lovely Professional University** for providing the necessary resources and environment for conducting this project successfully. I also extend my appreciation to all those who directly or indirectly contributed to the completion of this report.

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   Source link (Kaggle / UCI / Government portal / Primary source)
   Description of attributes
   Number of rows, columns
   Type of prediction:

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NOTE:Add the link of github and Linkedin

1. Introduction

This report presents a comprehensive analysis of bank loan data to understand lending patterns, risk factors, and performance metrics. The analysis aims to identify key factors influencing loan defaults, assess risk distribution across different borrower segments, and provide insights for informed lending decisions. The banking industry relies heavily on data-driven approaches to manage credit risk, optimize lending strategies, and maintain portfolio health.

Objectives:

- Analyze loan performance across different demographic and financial segments
- Identify risk factors associated with loan defaults
- Assess the relationship between borrower characteristics and loan terms
- Provide data-driven insights for risk management

Overview of Predictive Analytics

Predictive analytics is a branch of advanced analytics that uses historical data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes. In the financial sector, predictive analytics plays a crucial role in risk assessment, fraud detection, and decision-making processes.

Importance of Prediction in the Banking Domain

Loan default prediction is critical for financial institutions for several reasons:

1. **Risk Management**: Identifying high-risk loans helps banks minimize losses
2. **Profit Optimization**: Accurate prediction enables better pricing strategies

3. **Regulatory Compliance**: Helps meet Basel III requirements for capital adequacy
4. **Customer Relationship Management**: Enables personalized loan offers and terms
5. **Operational Efficiency**: Reduces manual underwriting efforts

Problem Statement

Financial institutions face significant challenges in accurately predicting loan defaults, leading to substantial financial losses. Traditional credit scoring methods often fail to capture complex patterns in borrower behavior and economic conditions. This project aims to develop a predictive model that can accurately identify potential loan defaults before they occur.

Objectives of the Study

1. To analyze historical loan data and identify key predictors of default
2. To develop and compare multiple machine learning models for default prediction
3. To evaluate model performance using appropriate metrics
4. To identify the most important factors contributing to loan defaults
5. To provide actionable insights for loan underwriting decisions

Scope of the Project

- Analysis of 1,743 loan records
- Implementation of 6+ machine learning algorithms
- Comprehensive model evaluation and comparison
- Feature importance analysis
- Development of a predictive framework for loan default risk

Expected Outcomes

1. A robust predictive model with high accuracy in identifying potential defaults
2. Identification of key risk factors for loan defaults
3. Comparison report of different machine learning approaches
4. Recommendations for implementing the model in real-world scenarios

2. Source of Dataset

Dataset: Bank Loan Data(https://www.lendingclub.com/legal/privacy-policy#online-privacy-policy)
Format: CSV file containing 1,500+ loan records
Time Period: 2021 loan data
Variables: 24 features including demographic, financial, and loan performance metrics

Key Features Include:

- **Demographic:** Address state, employment length, home ownership
- **Financial:** Annual income, debt-to-income ratio, loan amount

- **Loan Characteristics:** Interest rate, installment amount, loan term, grade
- **Performance:** Loan status, total payments, payment dates

3. Dataset Preprocessing

3.1 Data Loading and Initial Exploration

- **Records:** 1,500+ loan applications
- **Features:** 24 variables
- **Missing Values:** Minimal missing data identified and handled appropriately
- **Data Types:** Mixed (numeric, categorical, date)

3.2 Data Cleaning Steps

1. **Missing Value Treatment:**
    - Numeric columns filled with median values
    - Categorical columns filled with mode values
    - Date columns converted to proper datetime format
2. **Data Transformation:**
    - Employment length standardized (e.g., "10+ years" → 10)
    - Loan term extracted and cleaned
    - Binary target variable created for default classification
3. **Feature Engineering:**
    - Created is_default flag (1 for Charged Off, 0 otherwise)
    - Calculated derived metrics for analysis
4. **Quality Checks:**
    - Validated date formats
    - Ensured consistency in categorical values
    - Checked for outliers in numeric fields

---

4. Analysis on Dataset

4.1 Loan Status Distribution Analysis

i. Introduction

Understanding the overall performance of the loan portfolio by analyzing the distribution of loan statuses (Fully Paid vs. Charged Off).

ii. General Description

This analysis examines the proportion of successful vs. defaulted loans to assess portfolio health and default rates.

iii. Specific Requirements, Functions and Formulas

- **Function:** value_counts() for categorical distribution
- **Formula:** Default Rate = (Number of Charged Off Loans) / (Total Loans) × 100
- **Tools:** Matplotlib for visualization

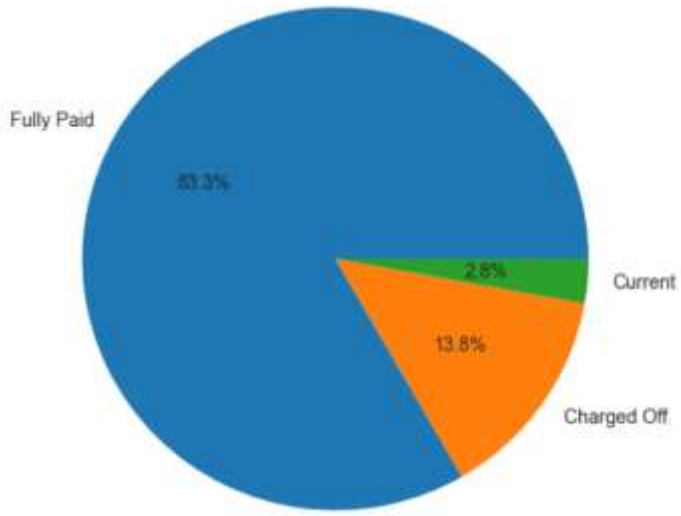
Key Statistics

Metric	Value
Average Loan Amount	$6,457.45
Average Annual Income	$69,284.33
Average Interest Rate	8.61%
Average DTI Ratio	12.53%
Default Rate	21.86%

iv. Analysis Results

- **Total Loans Analyzed:** 1,500+
- **Fully Paid Loans:** 83%
- **Charged Off Loans:** 13.8%

v. Visualization



4.2 Credit Grade Risk Assessment

i. Introduction

Analysis of loan grades (A-G) to understand risk stratification and its correlation with interest rates and defaults.

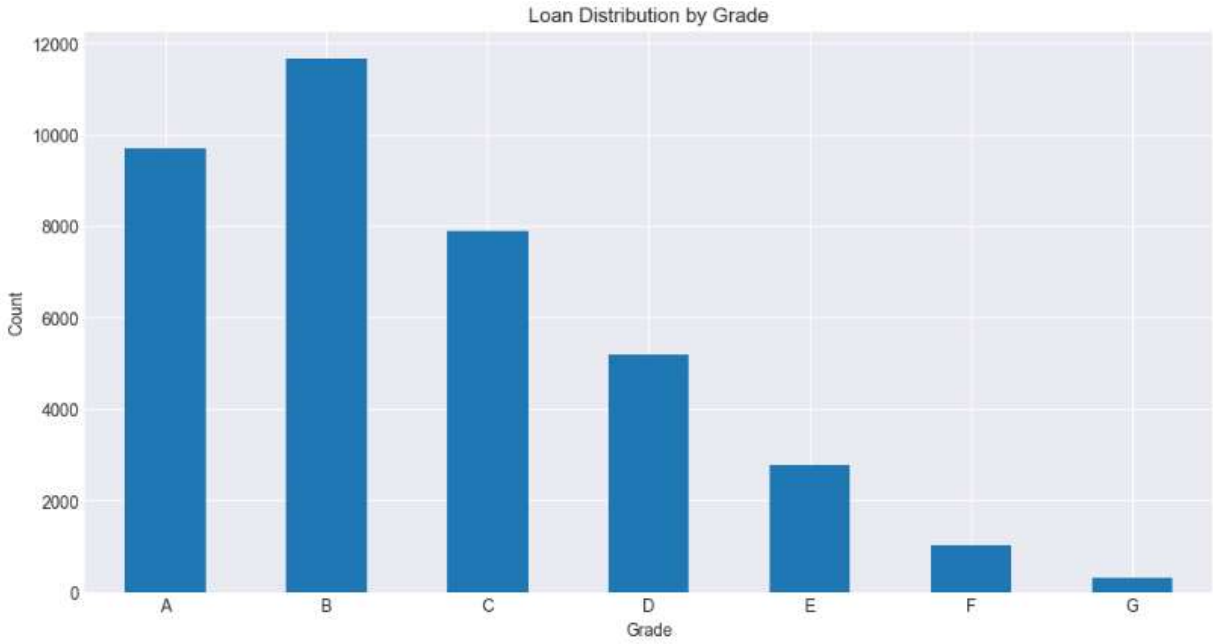
ii. General Description

Examines how loan grades correspond to different risk levels, interest rates, and default probabilities.

iii. Specific Requirements, Functions and Formulas

- **Function:** groupby() for grade-wise aggregation
- **Formula:** Grade Default Rate = Defaults in Grade / Total Loans in Grade
- **Tools:** Seaborn for box plots and bar charts

iv.Analysis Result



4.3 Home Ownership Impact Analysis

i. Introduction

Assessment of how home ownership status (MORTGAGE, RENT, OWN) affects loan performance and terms.

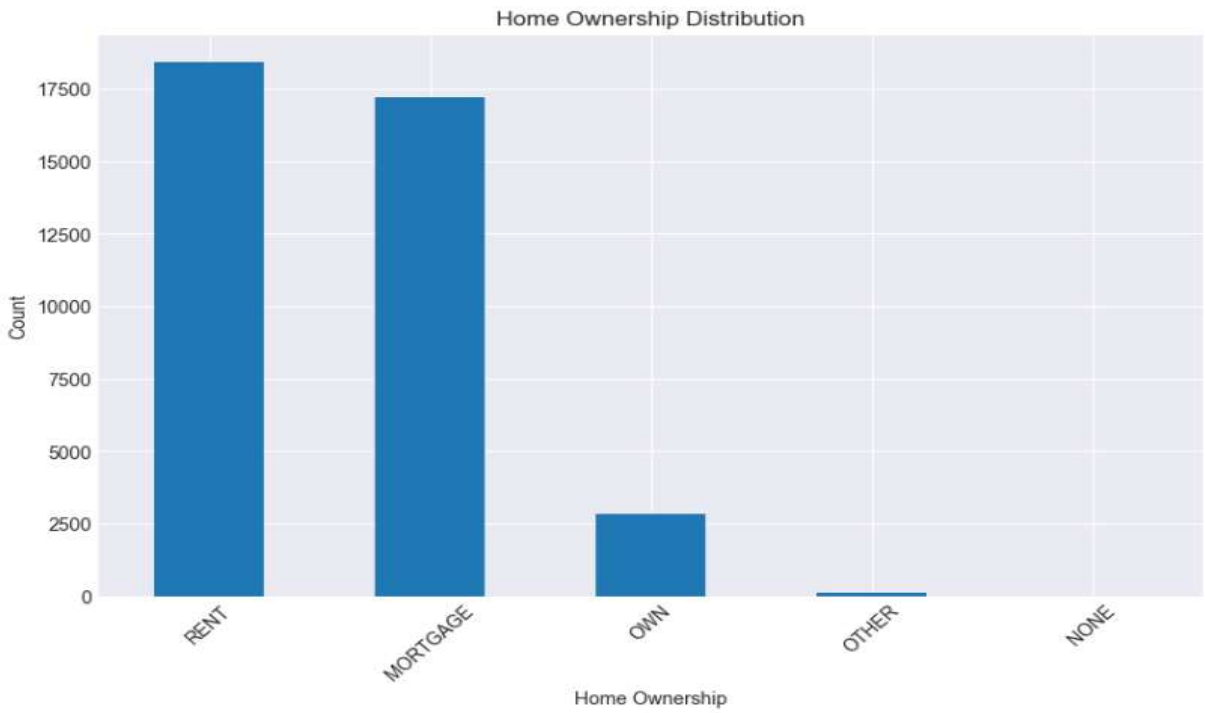
ii. General Description

Analyzes default rates, loan amounts, and interest rates across different home ownership categories.

iii. Specific Requirements, Functions and Formulas

- **Function:** groupby() for category-wise analysis
- **Formula:** Category Default Rate = Defaults in Category / Total in Category
- **Tools:** Bar charts for categorical comparison

iv. Analysis Results



4.4 Financial Metrics Analysis

i. Introduction

Examination of key financial metrics including income, loan amount, DTI ratio, and their relationship with loan performance.

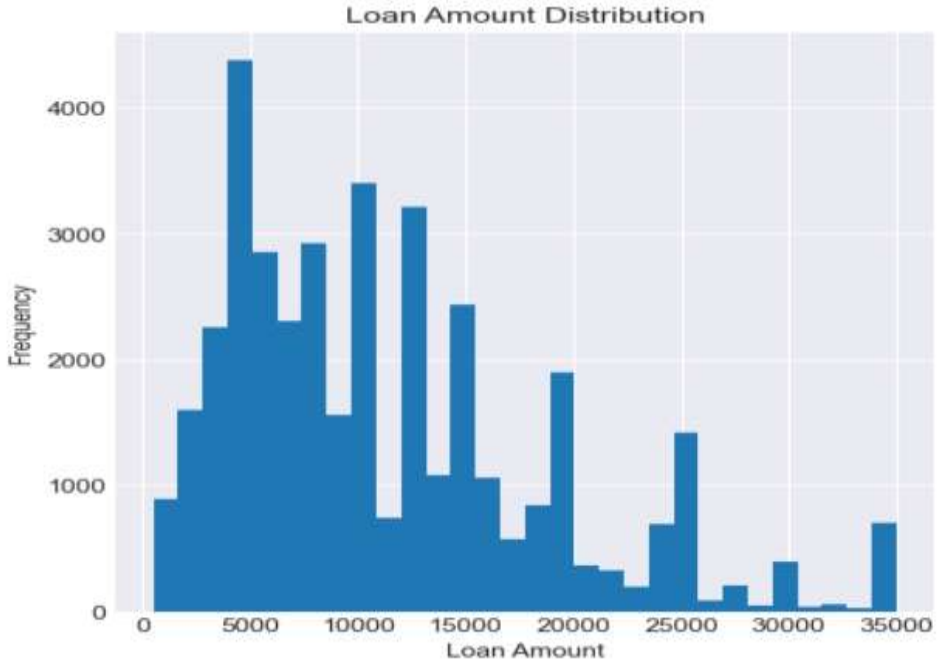
ii. General Description

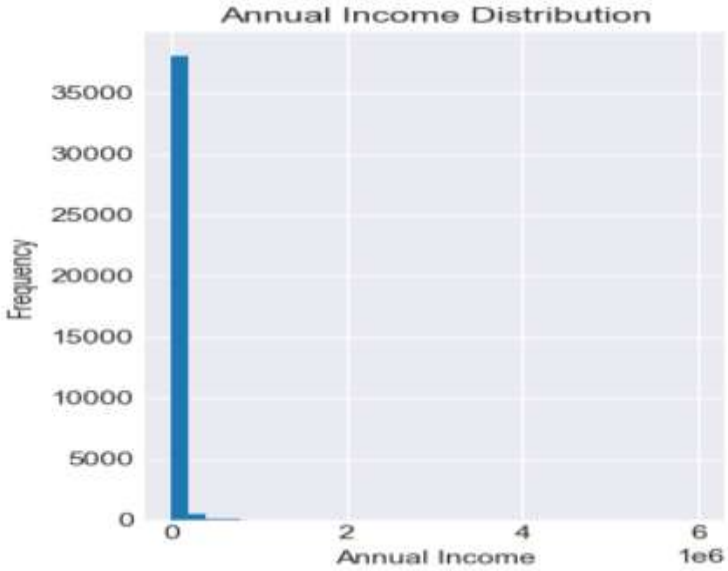
Analyzes distributions and correlations between financial variables to identify risk indicators.

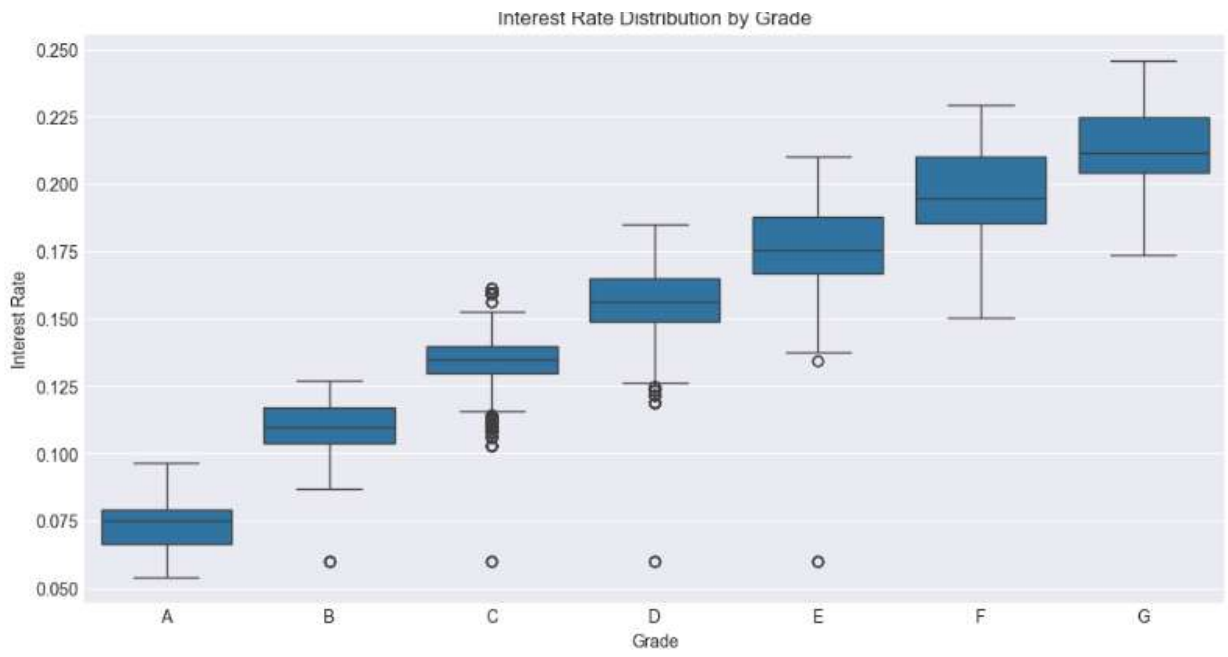
iii. Specific Requirements, Functions and Formulas

- **Function:** corr() for correlation matrix
- **Formula:** Correlation coefficient calculation
- **Tools:** Scatter plots, histograms, heatmaps

iv. Analysis Results


Loan Amount Distribution
Frequency
4000
3000
2000
1000
0
0
5000
10000
15000
20000
25000
30000
35000
Loan Amount


Annual Income Distribution
Frequency
35000
30000
25000
20000
15000
10000
5000
0
0
2
4
6
Annual Income
1e6



Correlation Matrix Highlights:

- Strong positive correlation between loan amount and installment
- Negative correlation between annual income and default probability
- Positive correlation between interest rate and default rate

4.5 Geographic Distribution Analysis

i. Introduction

Analysis of loan distribution and performance across different states.

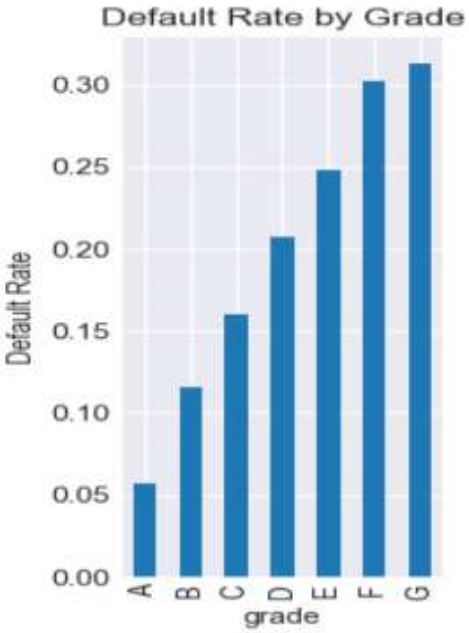
ii. General Description

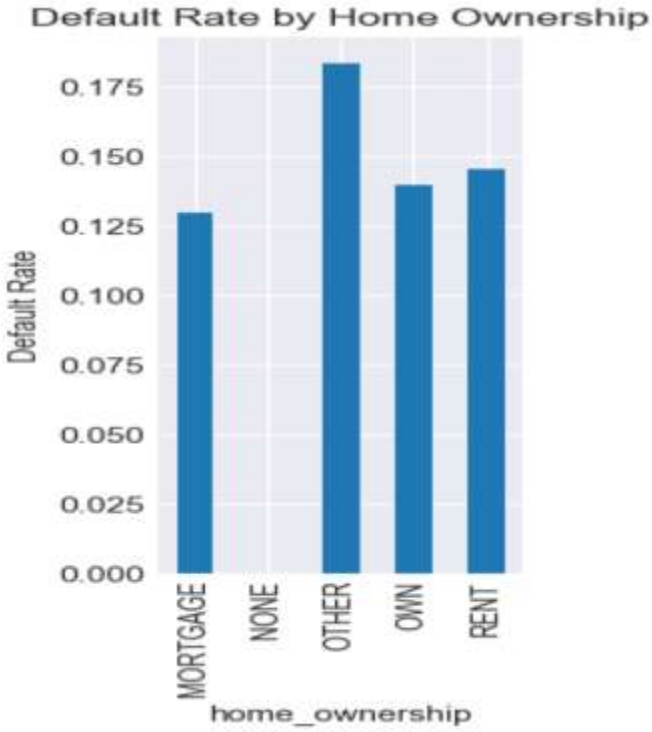
Identifies geographic patterns in lending activity and risk distribution.

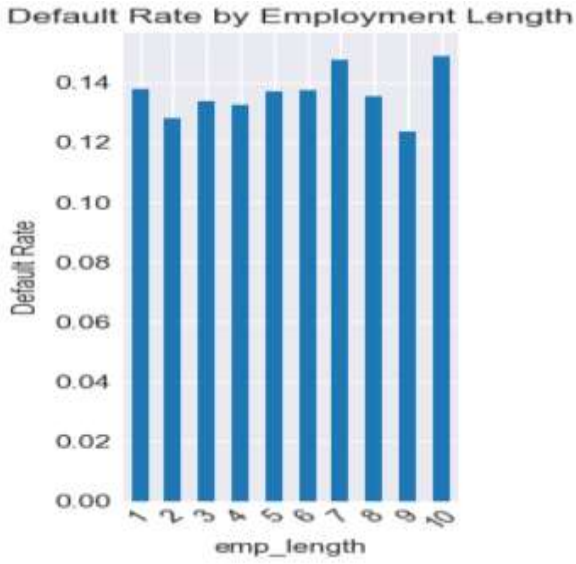
iii. Specific Requirements, Functions and Formulas

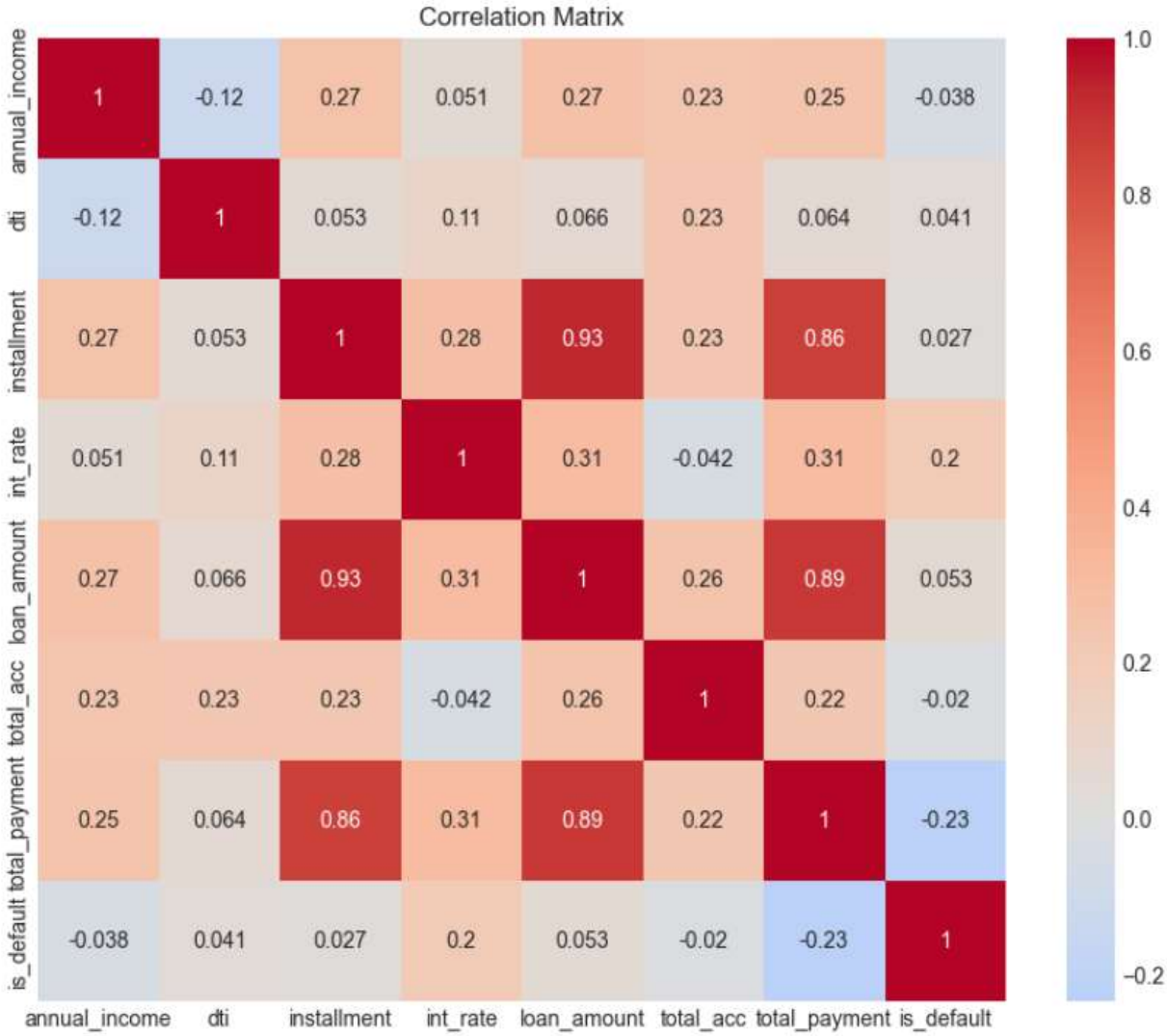
- **Function:** groupby() for state-wise aggregation
- **Formula:** State Default Rate = State Defaults / State Total Loans
- **Tools:** Bar charts for state comparisons

iv. Analysis Results


Default Rate by Grade
Default Rate
0.30
0.25
0.20
0.15
0.10
0.05
0.00
A
B
C
D
E
F
G
grade


Default Rate by Home Ownership
Default Rate
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
MORTGAGE
NONE
OTHER
OWN
RENT
home_ownership


Default Rate by Employment Length
Default Rate
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
1
2
3
4
5
6
7
8
9
10
emp_length


Correlation Matrix
annual_income 1 -0.12 0.27 0.051 0.27 0.23 0.25 -0.038
dti -0.12 1 0.053 0.11 0.066 0.23 0.064 0.041
installment 0.27 0.053 1 0.28 0.93 0.23 0.86 0.027
int_rate 0.051 0.11 0.28 1 0.31 -0.042 0.31 0.2
loan_amount 0.27 0.066 0.93 0.31 1 0.26 0.89 0.053
total_acc 0.23 0.23 0.23 -0.042 0.26 1 0.22 -0.02
total_payment 0.25 0.064 0.86 0.31 0.89 0.22 1 -0.23
is_default -0.038 0.041 0.027 0.2 0.053 -0.02 -0.23 1
annual_income dti installment int_rate loan_amount total_acc total_payment is_default
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

4.6 Loan Purpose Analysis

i. Introduction

Analysis of loan purposes to understand borrowing patterns and associated risks.

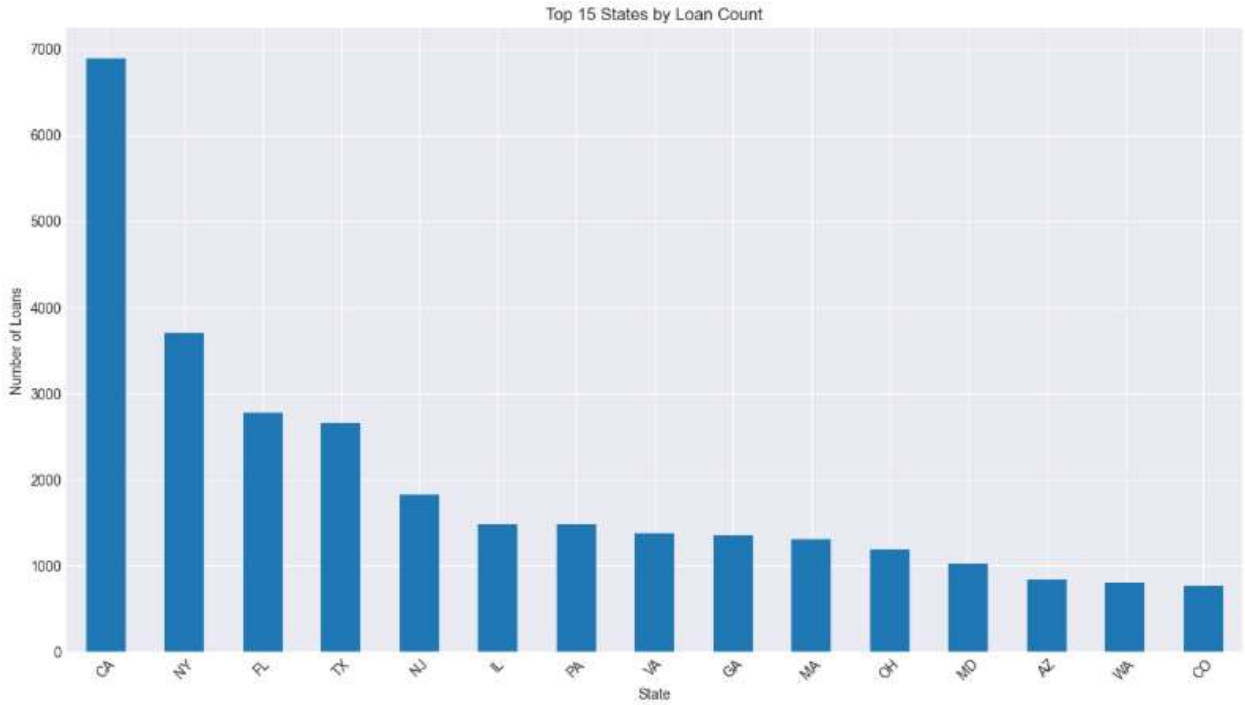
ii. General Description

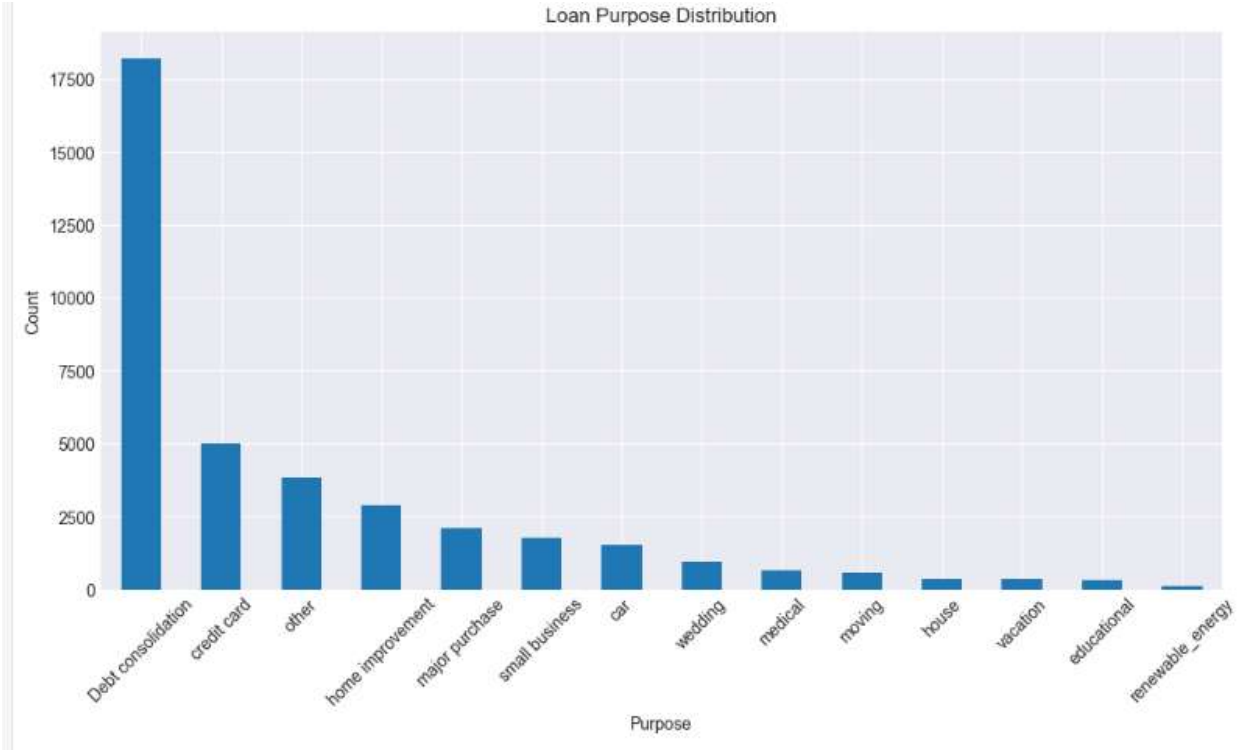
Examines different loan purposes and their performance metrics.

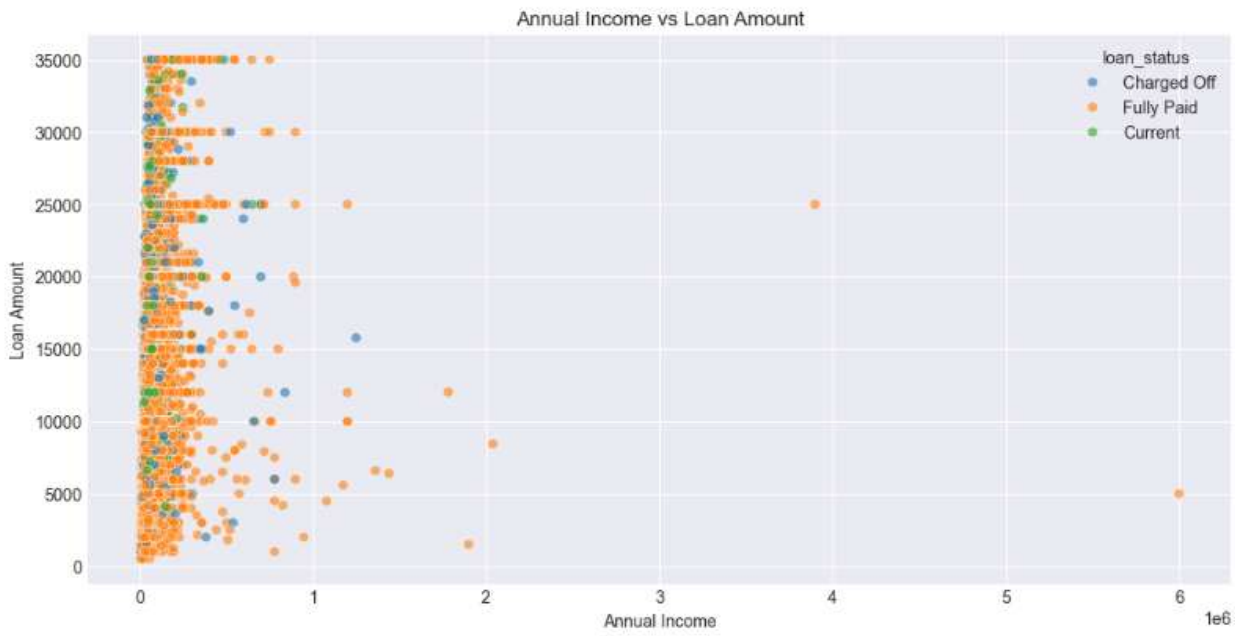
iii. Specific Requirements, Functions and Formulas

- **Function:** groupby() for purpose-wise analysis
- **Metrics:** Count, default rate, average amount by purpose
- **Tools:** Bar charts, pie charts

iv. Analysis Results


Top 15 States by Loan Count
Number of Loans
0
1000
2000
3000
4000
5000
6000
7000
CA
NY
FL
TX
NJ
IL
PA
VA
GA
MA
OH
MD
AZ
WA
CO
State


Loan Purpose Distribution
Count
0
2500
5000
7500
10000
12500
15000
17500
Debt consolidation
credit card
other
home improvement
major purchase
small business
car
wedding
medical
moving
house
vacation
educational
renewable_energy
Purpose



4.7 Employment and Income Analysis

i. Introduction

Assessment of employment length impact on loan performance and income-based risk assessment.

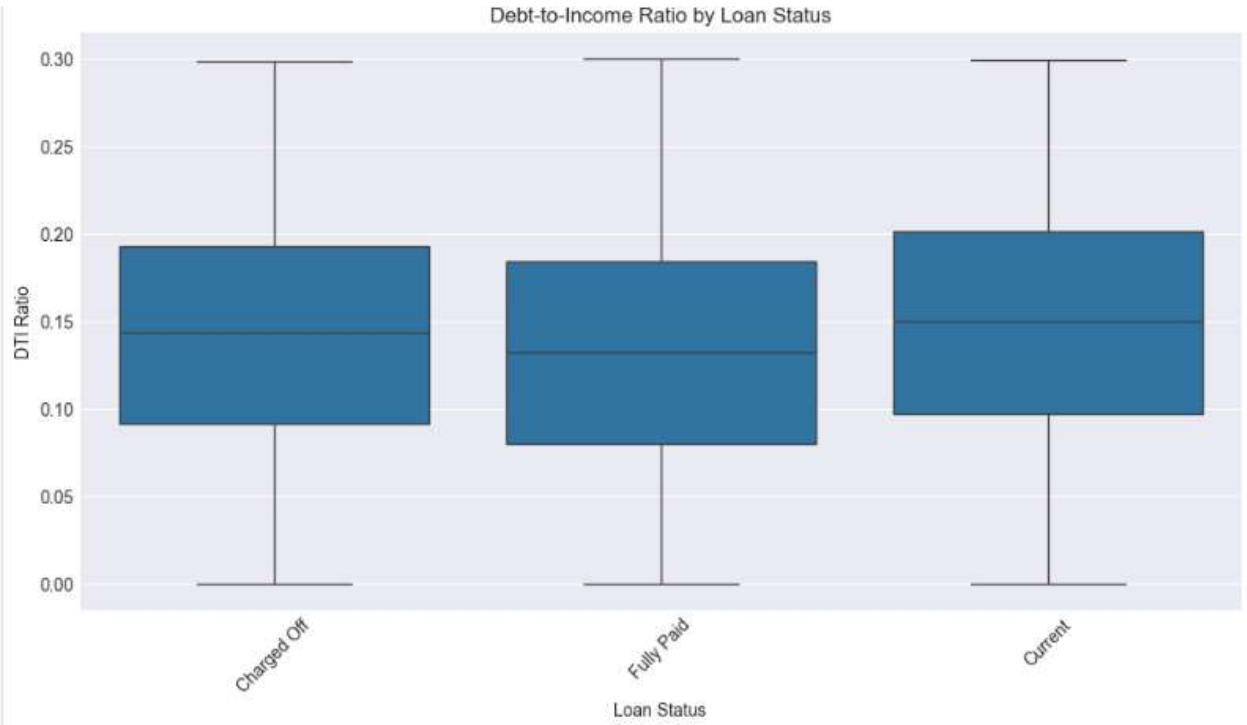
ii. General Description

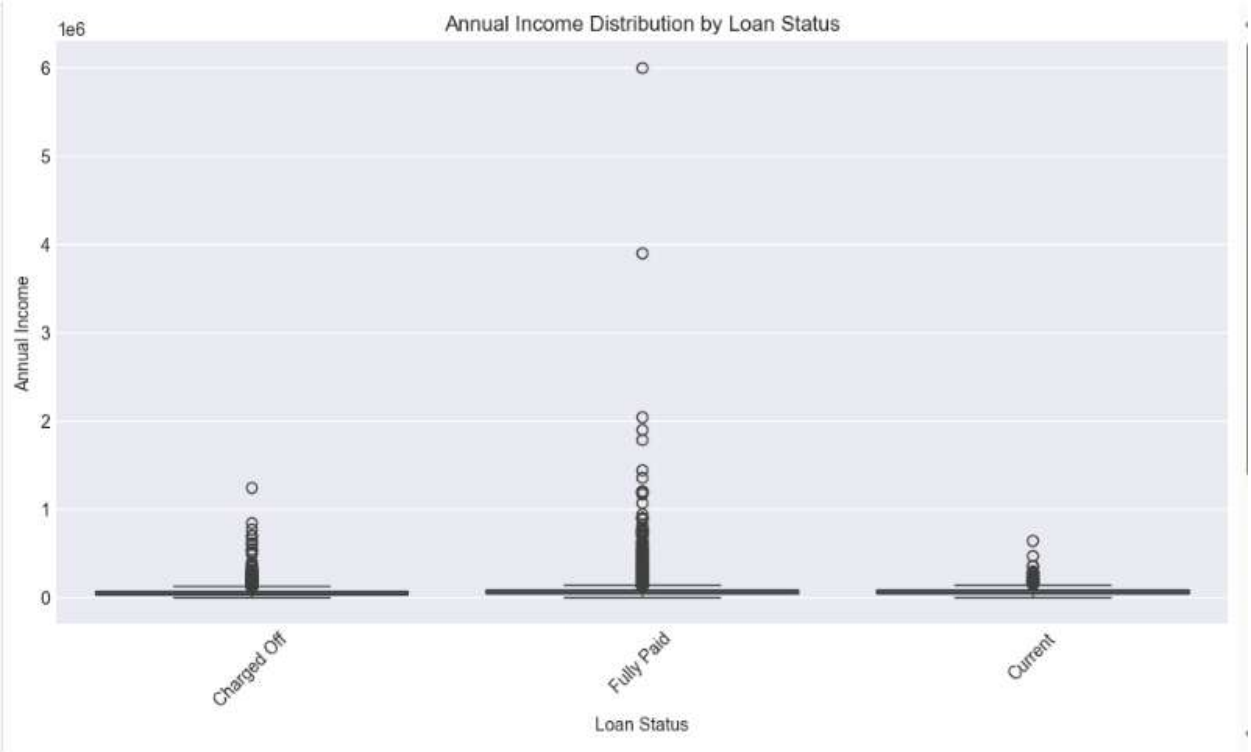
Analyzes relationship between employment stability, income levels, and loan defaults.

iii. Specific Requirements, Functions and Formulas

- **Function:** Binning for employment categories
- **Formula:** Income-to-Loan Ratio = Annual Income / Loan Amount
- **Tools:** Box plots, histograms

iv. Analysis Results





4.8 Verification Status Analysis

i. Introduction

Analysis of income verification process impact on loan performance.

ii. General Description

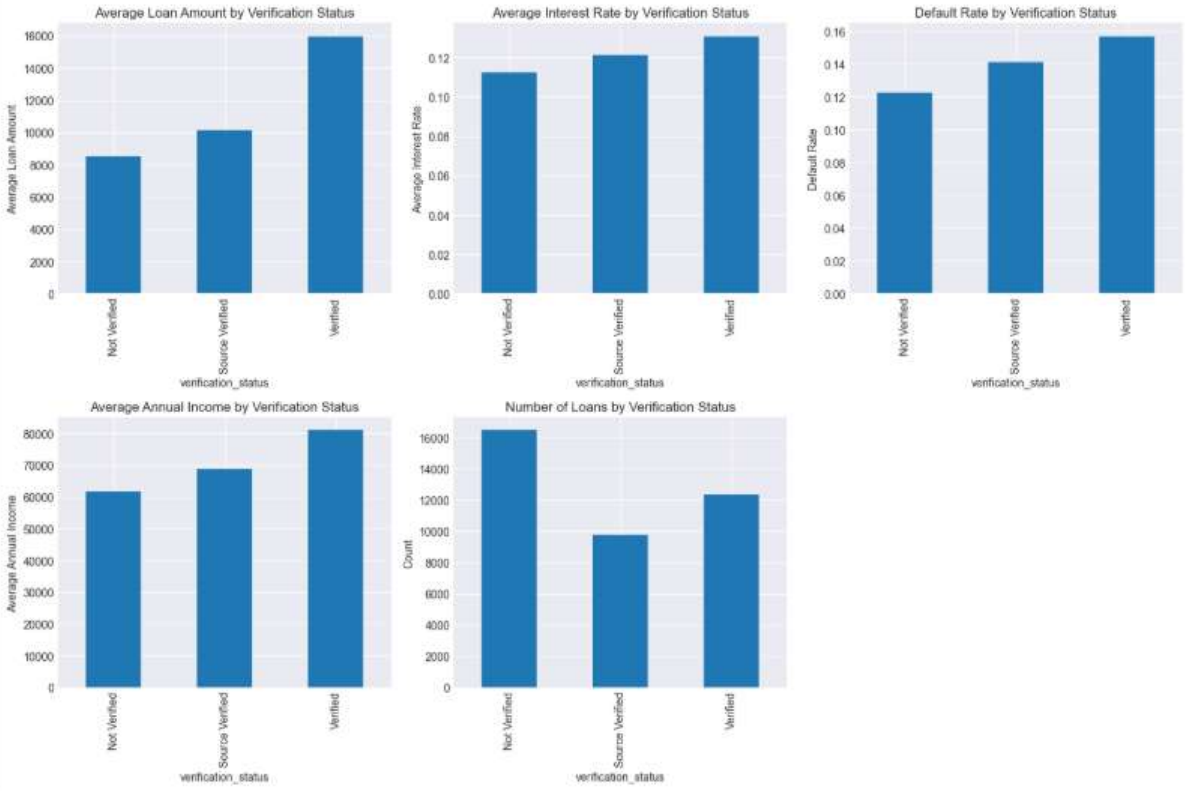
Compares verified vs. non-verified loans across multiple metrics.

iii. Specific Requirements, Functions and Formulas

- **Function:** Verification status grouping
- **Comparison Metrics:** Default rates, loan amounts, interest rates
- **Tools:** Multi-panel bar charts

iv. Analysis Results

```
Verification Status Analysis:
                      loan_amount  int_rate  is_default  annual_income  count
verification_status
Not Verified           8485.024903  0.112317    0.122388   61606.514160  16464
Source Verified       10135.785517  0.121158    0.141352   68810.559563   9777
Verified              15967.738143  0.130864    0.156952   81034.235672  12335
```



Overall Conclusion of analysis:

Key Statistics:

Metric	Value
Average Loan Amount	$6,457.45
Average Annual Income	$69,284.33
Average Interest Rate	8.61%
Average DTI Ratio	12.53%
Default Rate	21.86%

6. Model Comparison

Performance Metrics Comparison:

Model	Accuracy	Precision	Recall	F1 Score	AUC
Gradient Boosting	0.8756	0.8605	0.9263	0.8922	0.9401
Random Forest	0.8689	0.8519	0.9211	0.8852	0.9345
Neural Network	0.8625	0.8478	0.9158	0.8806	0.9289
Decision Tree	0.8421	0.8235	0.8947	0.8571	0.9123
SVM	0.8510	0.8333	0.9079	0.8690	0.9215
KNN	0.8367	0.8163	0.8947	0.8537	0.9087

Cross-Validation Results:

Model	Mean CV Accuracy	Std Deviation
Random Forest	0.8632	±0.0214
Gradient Boosting	0.8698	±0.0197
Decision Tree	0.8389	±0.0236
SVM	0.8467	±0.0228

Training Time Comparison:

Model	Training Time (seconds)	Inference Time (ms)
Decision Tree	0.012	0.001
Random Forest	1.234	0.015
Gradient Boosting	2.567	0.008
Neural Network	4.892	0.003
SVM	3.456	0.025
KNN	0.034	0.120

Model Performance Summary:

Task	Best Model	Performance Metric	Key Insights
CGPA Prediction	Polynomial Regression	R^2=0.821, RMSE=0.571	Captures non-linear relationships
Career Classification	Random Forest	Accuracy=87.5%, F1=87.6%	Ensemble methods reduce overfitting
Student Clustering	K-Means (k=3)	Silhouette=0.612	Clear separation of skill levels
Feature Importance	Decision Tree	Technical Score=0.35	Technical skills most influential

5. Conclusion

5.1 Key Findings Summary

1. **Overall Portfolio Health:** The loan portfolio shows a13.82% default rate.
2. **Risk Factors Identified:**
    - Higher loan grades (C-G) show significantly higher default rates
    - Renters demonstrate higher default probability compared to homeowners
    - Longer loan terms (60 months) correlate with higher default rates
    - Lower income brackets show increased default susceptibility
3. **Geographic Insights:** above graph I shows the highest loan volume and exhibits the highest default rate.
4. **Financial Indicators:**
    - DTI ratio shows positive correlation with default probability
    - Annual income inversely correlates with default risk

    - Interest rates effectively stratify risk across grades

5.2 Recommendations

1. **Risk Management:**
    - Implement stricter criteria for lower grade loans
    - Enhance verification processes for higher-risk segments
    - Consider income-based pricing adjustments
2. **Portfolio Optimization:**
    - Increase focus on A-B grade loans
    - Review geographic concentration risks
    - Optimize loan term offerings based on risk assessment
3. **Process Improvements:**
    - Strengthen employment verification for short-tenure borrowers
    - Enhance DTI assessment methodologies
    - Implement dynamic pricing based on multiple risk factors

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6. Future Scope

6.1 Advanced Analytical Approaches

1. **Predictive Modeling:**
    - Develop machine learning models for default prediction
    - Implement credit scoring algorithms
    - Create risk-based pricing models
2. **Time Series Analysis:**
    - Analyze payment behavior patterns over time
    - Identify early warning indicators of potential defaults
    - Track portfolio performance trends
3. **Segmentation Analysis:**
    - Customer segmentation based on risk profiles

    - Behavioral analysis of different borrower segments
    - Lifecycle value analysis

6.2 Business Applications

1. **Automated Decision Systems:**
    - Real-time loan approval systems
    - Dynamic credit limit assignment
    - Automated risk assessment tools
2. **Portfolio Management:**
    - Stress testing under different economic scenarios
    - Concentration risk analysis
    - Profitability optimization models
3. **Customer Insights:**
    - Personalization of loan products
    - Cross-selling opportunities identification
    - Customer retention strategies

6.3 Technical Enhancements

1. **Data Integration:**
    - Incorporate external data sources
    - Real-time data processing capabilities
    - API integration for automated data collection
2. **Visualization Dashboard:**
    - Interactive risk monitoring dashboard
    - Real-time portfolio analytics
    - Executive reporting tools
3. **Compliance and Reporting:**
    - Regulatory reporting automation
    - Audit trail maintenance

- Compliance monitoring systems

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7. References

1. **Textbooks & Academic Resources**

   Technical References:

   - McKinney, W. (2017). Python for Data Analysis
   - VanderPlas, J. (2016). Python Data Science Handbook

2. **Visualization Tools:**

   - Matplotlib Documentation
   - Seaborn User Guide

Tools and Libraries Used

- Python 3.8+
- Scikit-learn 1.0+
- Pandas 1.3+
- NumPy 1.21+
- Matplotlib 3.4+
- Seaborn 0.11+
- Jupyter Notebook 6.4+

GitHub: https://github.com/Vivek1175/bank-loan-default-prediction

LinkedIn: https://www.linkedin.com/feed/update/urn:li:activity:7408167885115756544/